Project Charter – Environment & Climate Analytics

Capstone Project – Project Charter

Group - Members

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Background

Environmental conditions across Atlantic Canada have demonstrated measurable changes over recent decades, driven by a combination of long-term climate change, anthropogenic activity, natural variability, and seasonal meteorological patterns. These changes manifest across interconnected domains, including average temperature trends, precipitation regimes, extreme weather event frequency, greenhouse gas (GHG) emissions, and air quality conditions.

Federal and provincial agencies—including Environment and Climate Change Canada (ECCC) and Climate Change Nova Scotia—publish extensive datasets covering historical climate observations, homogenized long-term climate records, emissions inventories, and air quality indicators. Despite the richness of these datasets, they remain distributed across platforms, formats, temporal frequencies, and reporting frameworks, making integrated environmental analysis challenging.

Assessing how average temperatures and precipitation patterns have evolved over the past 30 years, while simultaneously evaluating extreme weather event frequency, provincial GHG emission changes, and air quality trends, requires reconciling multiple data sources such as station-level climate observations, long-term climate baselines, emissions inventories, and meteorology-influenced AQI measurements. Currently, these factors are typically analyzed in isolation using static tables or siloed reporting tools, with limited support for anomaly detection, trend analysis, correlation exploration, or short-term forecasting across environmental domains.

This project proposes the development of an **Environment & Climate Analytics dashboard** that integrates authoritative public datasets into a unified Business Intelligence (BI) framework, enabling consistent analysis of long-term trends, variability, spatial patterns, and near-term insights for Atlantic Canada, with focused analysis on Nova Scotia.

Problem Statement

Although Canada's environmental and climate datasets are publicly available and scientifically robust, their fragmented structure, inconsistent granularity, and lack of integrated analytical tooling limit their practical use for data-driven analysis. Users face difficulty transforming raw environmental data into clean, well-modeled datasets, applying exploratory and predictive analytics, and communicating insights effectively to non-technical audiences. From a Business Intelligence and Data Analytics perspective, there is a need for an end-to-end analytics solution that consolidates environmental data, applies exploratory and predictive methods using tools such as Pandas, regression, and scikit-learn, and presents results through interactive Power BI dashboards. This project addresses that gap by demonstrating how BI workflows can convert complex environmental data into accessible, decision-ready insights that support analysis, learning, and informed discussion.

Goals

- Integrate provincial and federal environmental datasets into a **unified analytics model** using consistent spatial and temporal dimensions
- Analyze **30-year historical trends** in average temperature and precipitation across Atlantic Canada
- Identify and visualize **temperature and precipitation anomalies** relative to historical baselines
- Evaluate whether a **measurable trend exists in extreme weather event frequency** (high wind and heavy precipitation) at Nova Scotia weather stations
- Compare **GHG emissions changes** across Canadian provinces and identify the largest contributors by region and sector
- Analyze **Air Quality Index (AQI) trends**, seasonal exceedances, and meteorological correlations
- Develop a **5-year forecast** for average annual temperature at a selected Nova Scotia weather station
- Deliver interactive, insight-driven dashboards using **Power BI** that support exploratory and comparative analysis

Scope

The project will deliver an **interactive environment and climate analytics dashboard** focused on **Atlantic Canada**, with enhanced station-level detail for **Nova Scotia**. The analysis will integrate climate observations, greenhouse gas emissions (province wise, 10Y data), and air quality indicators (Canada) covering approximately **1995–2025**, subject to dataset availability.

The dashboard will support analysis of:

- Average annual temperature trends
- Annual precipitation patterns
- Extreme weather event frequency
- Provincial and sectoral GHG emissions
- Air Quality Index behavior and correlations
- Short-term temperature forecasting

All data will be sourced from **publicly available provincial and federal datasets** and modeled to ensure analytical consistency.

In Scope	Out of Scope
• Integration of historical climate datasets (temperature, precipitation, wind) • Use of **30-year climate baselines** for anomaly analysis	• Real-time or streaming environmental data • Long-term climate scenario projections (e.g., CMIP6 future scenarios)

• Station-level analysis for Nova Scotia weather stations • Extreme weather event identification and frequency trend assessment • Provincial and sector-level GHG emissions analysis • AQI trend, seasonal exceedance, and correlation analysis • Exploratory data analysis (EDA) using Python (Pandas) • Basic trend modeling (linear / polynomial regression) • Short-term forecasting using time-series methods (e.g., ARIMA) • Export of anomaly and forecast outputs for Power BI • Interactive dashboard development (KPIs, trends, maps, slicers)	• Policy evaluation or emissions reduction recommendation frameworks • Health outcome modeling or impact assessments • Production-grade automated data pipelines or APIs • Custom data collection or sensor deployment

Outcome Expectations

The project will result in a **data-driven analytics dashboard** that provides clear, interpretable insights into **environmental and climate patterns across Atlantic Canada**.

Key outcomes include:

- Clear visualization of long-term climate trends and anomalies
- Evidence-based assessment of extreme weather event frequency
- Identification of provinces and sectors driving GHG emissions change
- Improved understanding of AQI variability and atmospheric influences
- A defensible short-term temperature forecast for Nova Scotia
- Improved environmental data literacy through interactive visual storytelling

Success Criteria

- Public environmental datasets are successfully cleaned, standardized, and integrated into a unified analytical model.
- Core analytical questions related to climate trends, extreme weather, GHG emissions, air quality, and forecasting are clearly addressed.
- Business Intelligence and Data Analytics tools (Pandas, regression/scikit-learn, Power BI) are applied appropriately and effectively.

- Interactive and user-friendly Power BI dashboards with meaningful KPIs and visualizations are delivered.
- Analytical results are validated, with assumptions, limitations, and data definitions clearly documented.
- All project milestones, checkpoints, and final deliverables are completed according to the approved timeline.
- Final outputs meet NSCC academic and professional standards for analysis, documentation, and presentation.

Key Stakeholders

Role	Name
Project Client & Sponsor and Subject Matter Expert	Patrick Dolinger (Faculty, NSCC)
Project Manager & QA	Yashaswi Tiwari (Student, NSCC)
Project Team Members	
• SQL/ETL Analyst	Chisom Njoku (Student, NSCC)
• Data Lead & SQL/ETL Analyst	Danielle Aranha (Student, NSCC)
• Reporting Lead & SQL/ETL Analyst	Michael Okafor (Student, NSCC)
• Documentation	Tracy Odaro (Student, NSCC)

Stakeholder Personas

The Project Sponsor & Subject Matter Expert (Sponsor Patrick)

- **Profile:** Patrick Dolinger is a faculty member at NSCC providing academic oversight and subject matter expertise for the project. He ensures methodological rigor, alignment with program outcomes, and adherence to professional BI standards.
- **Goal:** To guide students in applying sound Business Intelligence and data analytics practices to real-world environmental datasets while ensuring project scope, quality, and academic expectations are met.
- **BI Value:** Patrick uses the dashboards and documented analytical outputs to evaluate methodological correctness, validate assumptions, and ensure insights are defensible, enabling effective knowledge transfer and assessment without direct involvement in data preparation.

Other Stakeholders (Data Consumers): Environmental analysts, planners, policy analysts, students, and educators who use the dashboards and insights for analysis, planning, reporting, and learning purposes.

1. The Policy Planner (“Policy-Maker Paula")

- **Profile:** Government official focused on regional environmental compliance.

- **Goal:** To identify which sectors in Nova Scotia are the primary drivers of GHG emissions to inform new regulations.
- **BI Value:** She uses the **sector-level GHG charts** to get high-level insights for provincial reports without needing to clean raw data herself.

2. The Academic Researcher ("Scholar Sam")

- **Profile:** University student or researcher analyzing long-term climate shifts.
- **Goal:** To validate whether extreme weather events (high wind/rain) have actually increased in frequency over the last 30 years.
- **BI Value:** He relies on the **homogenized climate baselines** and **anomaly visualizations** to prove historical trends with scientifically robust data.

3. The Public Health Advocate ("Analyst Alex")

- **Profile:** Analyst for a non-profit interested in air quality and community well-being.
- **Goal:** To understand the correlation between meteorological patterns and seasonal Air Quality Index (AQI) spikes.
- **BI Value:** Uses **Power BI slicers** to compare seasonal AQI exceedances against temperature and precipitation trends to issue community health warnings.

Project Duration

The project will run from **April 22, 2026 to May 22, 2026**, following a structured academic timeline comprising:

Week / Dates	Milestone / Phase	Key Work & Deliverables
Week 1 (22–24 Apr)	Scoping, Charter & Data Acquisition	Team formation and role assignment. Project charter drafted. Dataset review and acquisition. GitHub repository setup.
Week 2 (27 Apr–1 May)	Data Cleaning & ETL	SQL/Python data cleaning. Power Query ETL. Data dictionary and schema documentation. Checkpoint #1.
Week 3 (4–8 May)	Exploratory Analysis & Predictive Modelling	Pandas EDA. Analytical questions defined. Predictive modelling. Validation and export. Checkpoint #2.
Week 4 (11–15 May)	Power BI Dashboard Build & Iteration	Initial Power BI build. DAX measures and interactivity. QA and stakeholder review. Checkpoint #3.

Week 5 (18–22 May)	Final Delivery & Project Closure	Final dashboard polish. User manual. Presentation rehearsal. Final submission and presentation.

Proposed Data Sets

- **Nova Scotia Open Data Portal** – Historical climate observations
- **ECCC Geo Network (AHCCD)** – Homogenized climate data for temperature and precipitation
- **Government of Canada Open Data** – Historical weather and station metadata
- **ECCC Environmental Indicators – GHG Emissions** – Provincial and sectoral emissions
- **ECCC Environmental Indicators – Air Quality (AQI)** – Station-level air quality data
- **Canadian Environmental Sustainability Indicators (CESI)** – Indicator definitions and context

Constraints, Assumptions, and Risks

Constraints

- Reliance on publicly available datasets
- Variation in spatial and temporal granularity across sources
- Limited to free analytics tools (Python, Power BI)

Assumptions

- Public datasets remain accessible and stable
- NSCC-provided tools are sufficient for modeling and visualization

Risks

- Data quality inconsistencies or missing observations
- Time constraints due to fixed academic schedule
- Limited forecasting accuracy due to simplified models

Approval Signatures

Name: **Patrick Dolinger** For Business Intelligence and Data Analytics program (NSCC) **Project Client**	Name: **Patrick Dolinger** Faculty, Technology and Environment, (NSCC) **Project Sponsor**	Name: **Yashaswi Tiwari** Business Intelligence and Data Analytics program Student, NSCC **Project Manager**